

Othmane Echchabi

M.Sc. Computer Science, McGill University & Mila

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RESEARCH INTERESTS

Geospatial foundation models; self-supervised and temporal representation learning; multimodal fusion for Earth observation, with applications to climate risk, wildfire, and food security.

EDUCATION

Aug. 2025 – 2027	M.Sc. Computer Science , McGill University & Mila – Quebec AI Institute. GPA 4.0/4.0. Advised by Prof. David Rolnick.
Aug. 2021 – May 2025	B.S. Data Science , Duke University / Duke Kunshan University. Thesis on self-supervised monitoring of <i>Spartina alterniflora</i> .

PUBLICATIONS

1. Y. Zhang*, **O. Echchabi***, T. Feng, W. Zhang, H.-K. Liao, C. Chang. Detecting Transportation Mode Using Dense Smartphone GPS Trajectories and Transformer Models. *International Journal of Geographical Information Science* (2026). [Paper](#) | [Code](#).
2. **O. Echchabi***, A. Lahlou*, N. Talty, J. M. Manto, T. Zheng, K. L. Lam. Seeing SDG 6 from Space: Local-Scale Monitoring of Piped Water and Sewage System Access Across Africa Using Satellite Imagery and Self-Supervised Learning. *Water Research X* 100600 (2026). [Paper](#).

*Equal contribution.

RESEARCH EXPERIENCE

Jan. 2026 – Present	Graduate Researcher , Mila – Quebec AI Institute, with the Allen Institute for AI (Ai2). <i>Geospatial foundation models for high-temporal-resolution tasks</i> .
Jul. – Dec. 2025	ML Research Intern , University Mohammed VI Polytechnic, Morocco. <i>Transformer and retrieval models for crop-yield recommendation</i> .
Jun. 2024 – Jun. 2025	Research Assistant , Duke Kunshan University, China. <i>Self-supervised satellite imagery; transformers for mobility</i> . With Dr. K. L. Lam, Dr. C. Chang.
Jan. – Jun. 2025	Research Assistant , University of Missouri. <i>Perceiver IO and VAE simulation for wildfire spread</i> . With Dr. J. Malof.
Jun. 2023 – Aug. 2024	Research Fellow , Duke University. <i>Vision transformers for coastal ecosystem and sea-level-rise mapping</i> . With Dr. W. Li, Dr. E. Bernhardt.

PRESENTATIONS

2024	Monitoring <i>Spartina alterniflora</i> Using Self-Supervised Learning . Poster, Duke Climate+ Symposium, Durham, NC.
2023	Assessing Climate Change Risk of Rural Coastal Plains . Poster, Duke Climate+ Symposium, Durham, NC.

GRANTS & AWARDS

2026–2028	Dan Stefan Petrescu Fellowship, McGill University – \$30,000/year for two years
2025–2027	McGill Computer Science Graduate Award (\$30,000/year) and Graduate Registration Bonus (\$5,000, one-time)
2023–2024	DKU Student Experiential Learning Fellowship (\$1,000); Signature Work Research Grant (\$1,500); Innovation Incubator (\$3,000)

TEACHING & SERVICE

2022–Present	Teaching Assistant , COMP 303 Software Design, McGill University; Mathematics & CS, Duke Kunshan University
2025–Present	Lab Representative , Mila – Quebec AI Institute; Board Member , Muslims in Tech
2022–2025	Software Developer , DKU Computer Science Club (HackDKU 2025); Resident Assistant and Varsity Soccer Co-Captain, DKU

PROFESSIONAL EXPERIENCE

Oct. – Nov. 2022	Data Analyst Intern , Atos Morocco, Rabat. <i>Real-time HR dashboards deployed across African branches</i> .
Jul. – Aug. 2022	Data Analyst Intern , XPerlean, Saint-Quentin, France. <i>Faster R-CNN and YOLO for ceramic defect detection</i> .
Oct. – Dec. 2021	Data Analyst Intern , Al Jazeera Media Institute, Doha. <i>Social-media data pipelines for engagement analysis</i> .

SKILLS

Programming: Python, SQL, Java, C/C++, JavaScript, R. **ML:** PyTorch, TensorFlow, scikit-learn, NumPy, pandas.
Geospatial: Google Earth Engine, ArcGIS, Sentinel-2/Landsat, xarray. **Systems:** Git, Docker, Linux, GCP, multi-GPU.
Languages: Arabic (native); French, English (fluent); Chinese (intermediate); Spanish (conversational).